



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

2023 Year 11 Assessment Task 2

Wednesday, 21st June 2023

General instructions

- Working time – 60 min.
(plus 5 minutes reading time)
- Write your answers in the space provided.
- Write using a blue or a black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.

Class (please ✓)

☐ 11MAA.1 – Miss Chaussivert

NESA STUDENT #

Marker's use only.

QUESTION	1-3	4-6	7-8	9-10	11-12	Total	%
MARKS	$\overline{7}$	$\overline{15}$	$\overline{8}$	$\overline{10}$	$\overline{7}$	$\overline{47}$	

Question 1 (2 marks)Find the exact value of $\tan 600^\circ$.**2**

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Question 2 (2 marks)

Find the solutions of:

$$\sin \theta = -\frac{\sqrt{3}}{2}$$

where $0^\circ \leq \theta \leq 360^\circ$.**2**

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Question 3 (3 marks)Solve the following for $0^\circ \leq x \leq 180^\circ$:

$$2 \cos^2 x - \cos x = 0$$

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Question 4 (6 marks)Solve the following equations for x :

(a) $3^{x-1} = 27$

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(b) $(\log_3 x)^2 = \log_3(x^5) - 6$

4

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Question 5 (3 marks)

Simplify the following, writing each pronumeral with a positive index.

3

$$\frac{\left(\sqrt{mn^{\frac{3}{2}}}\right)^4}{p^{-3}} \times (p^{-2})^{-3}$$

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Examination continues overleaf...

Question 6 (6 marks)

Consider the function $f(x) = \log_{10}(3x + 1)$.

- (a) State any vertical and horizontal asymptotes of $f(x)$. **1**

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- (b) Find all points where $f(x)$ intersects the coordinate axes. **2**

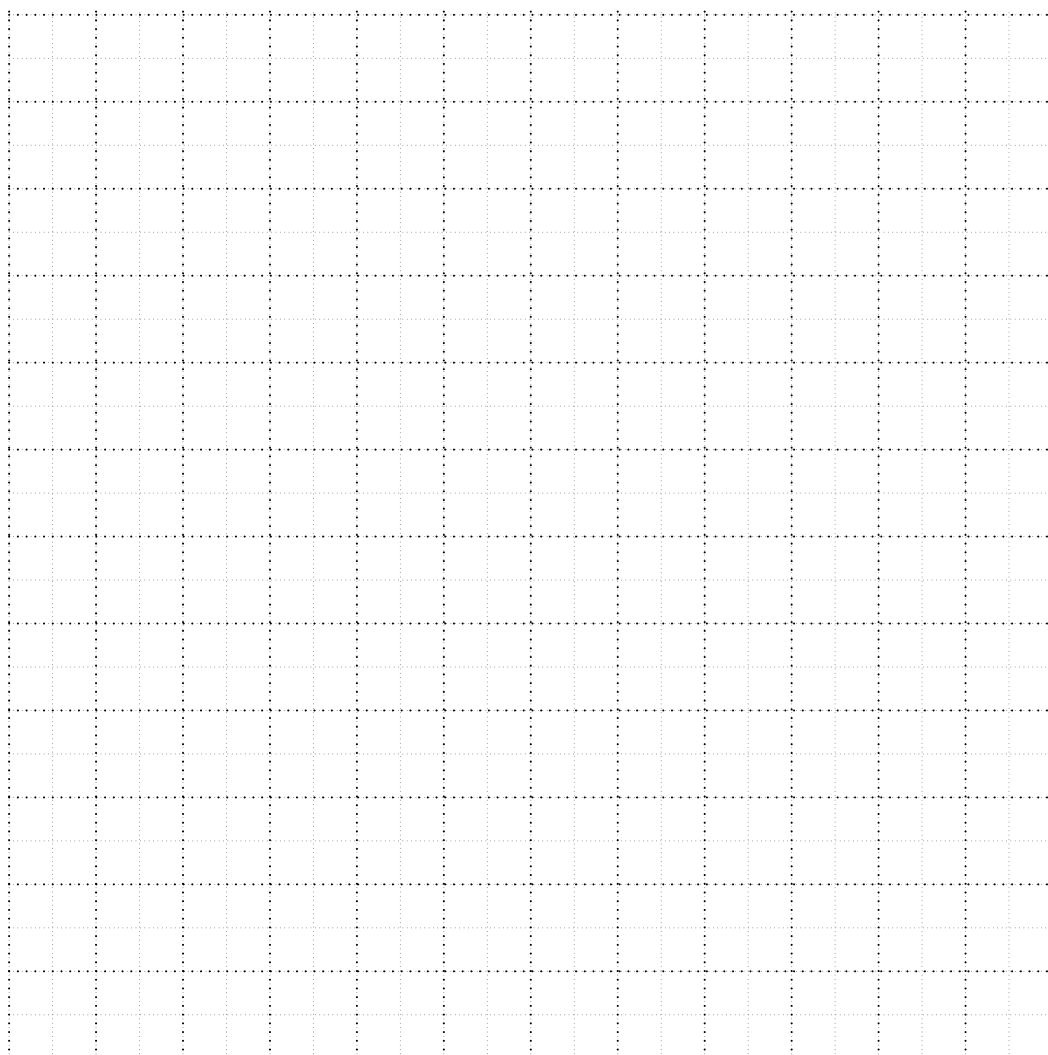
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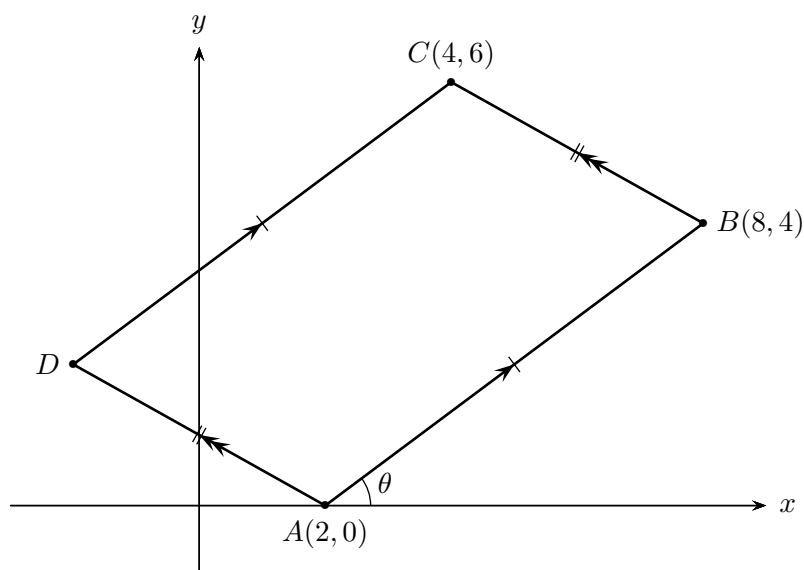
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- (c) Sketch the graph of $f(x)$, showing all important features. **3**



Question 7 (6 marks)

The points $A(2, 0)$, $B(8, 4)$, $C(4, 6)$ and D form the four vertices of a parallelogram, as shown in the diagram.



- (a) Find the gradient of the line AB and hence find the angle θ correct to the nearest minute. **2**

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.....

- (b) Show that the equation of line CD , parallel to AB and passing through C is given by $2x - 3y + 10 = 0$. **2**

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- (c) State the coordinates of point D . **2**

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Examination continues overleaf...

Question 8 (2 marks)**2**

Given the points $A(-1, 2)$, $B(3, 4)$, and $C(7, -4)$ in the Cartesian plane, show that $\triangle ABC$ is a right angled triangle.

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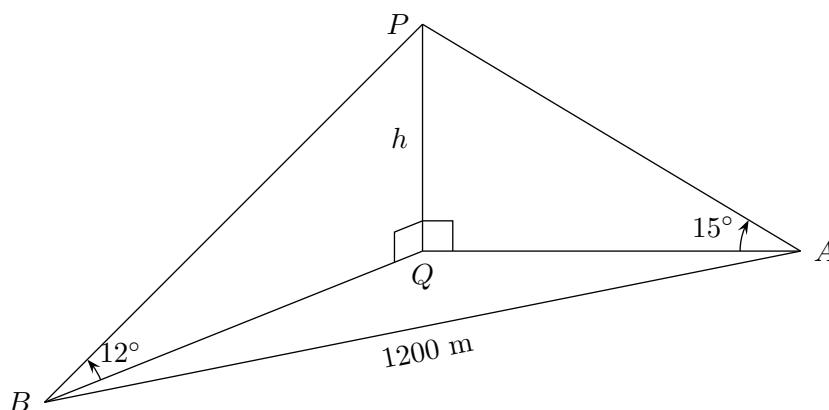
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Question 9 (7 marks)

A tower PQ of height h metres has an angle of elevation of 15° from a point A due east of it. From another point B , the bearing of the tower is 055°T and the angle of elevation is 12° . The points A and B are 1200 metres apart on the same level as the tower's base Q .



- (a) Show that $\angle AQB = 145^\circ$

1

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- (b) Consider the triangle APQ and show that $AQ = h \cot 15^\circ$.

1

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- (c) Find a similar expression for BQ .

1

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- (d) Show that $h = \frac{1200^2}{\cot 15^\circ \cot 12^\circ - 2 \cot 15^\circ \cot 12^\circ \cos 145^\circ}$ **3**

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- (e) Hence, calculate the value of h , correct to the nearest metre. **1**

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Question 10 (3 marks)

Given that $\cos \alpha = -\frac{2}{\sqrt{5}}$ and $180^\circ \leq \alpha \leq 360^\circ$, find the exact value of $\cot \alpha$.

3

Show all relevant working.

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Examination continues overleaf...

Question 11 (4 marks)

A newly bought car is worth \$10 000. After t years, the value of the car (in dollars) is given by the formula:

$$V = Ae^{-0.2t}$$

- (a) Find the value of A . **1**

.....

- (b) Find the value of the car after 6 years, correct to the nearest cent . **1**

.....

- (c) How many years (correct to 2 decimal places) will it take for the car's value to reach \$1 000? **2**

.....

Question 12 (3 marks)

Express:

$$\log(x^2\sqrt{y^3z})$$

in the form of $a \log x + b \log y + c \log z$.

.....

End of paper.



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Class (please ✓)

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NESA STUDENT #

Solutions

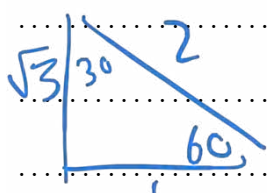
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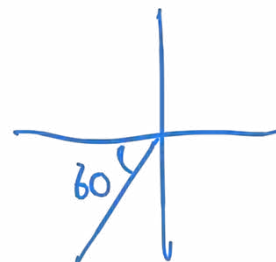
Question 1 (2 marks)Find the exact value of $\tan 600^\circ$.

2

$$\tan(600) = \tan(240) \checkmark$$



$$= \sqrt{3} \checkmark$$

**Question 2** (2 marks)

Find the solutions of:

2

$$\sin \theta = -\frac{\sqrt{3}}{2}$$

where $0^\circ \leq \theta \leq 360^\circ$.

3rd and 4th quadrant:
related angle: $60^\circ \checkmark$

$$\theta = 180 + 60, 360 - 60 = 240^\circ, 300^\circ \checkmark$$

Question 3 (3 marks)Solve the following for $0^\circ \leq x \leq 180^\circ$:

3

$$2 \cos^2 x - \cos x = 0$$

$$\cos x (2 \cos x - 1) = 0$$

$$\cos x = 0$$

$$\text{or } 2 \cos x - 1 = 0$$

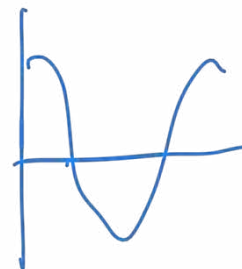
$$x = 90$$

$$2 \cos x = 1$$

$$\cos x = \frac{1}{2}$$

$$x = 60^\circ, 90^\circ \checkmark$$

$$x = 60^\circ$$



Question 4 (6 marks)Solve the following equations for x :

(a) $3^{x-1} = 27$

2

$$3^{x-1} = 3^3 \quad \checkmark$$

$$x-1 = 3$$

$$x = 4 \quad \checkmark$$

(b) $(\log_3 x)^2 = \log_3(x^5) - 6$

4

$$(\log_3 x)^2 = 5 \log_3 x - 6 \quad \checkmark$$

$$\text{let } u = \log_3 x \quad u^2 = 5u - 6 \quad \checkmark$$

$$u^2 - 5u + 6 = 0$$

$$(u-3)(u-2) = 0$$

$$u = 3, 2 \quad \checkmark \Rightarrow \log_3 x = 3 \quad \log_3 x = 2$$

$$x = 27, 9 \quad \checkmark$$

Question 5 (3 marks)

3

Simplify the following, writing each pronumeral with a positive index.

$$\frac{(\sqrt{mn^{\frac{3}{2}}})^{\frac{1}{2} \times 4}}{p^{-3}} \times (p^{-2})^{-3}$$

$$= \frac{m^2 n^3 p^6}{p^{-3}} \quad \checkmark$$

$$= m^2 n^3 p^9 \quad \checkmark$$

Examination continues overleaf...

Question 6 (6 marks)

Consider the function $f(x) = \log_{10}(3x + 1)$.

- (a) State any vertical and horizontal asymptotes of $f(x)$.

vertical asymptote: $x = -\frac{1}{3}$ ✓

1

- (b) Find all points where $f(x)$ intersects the coordinate axes.

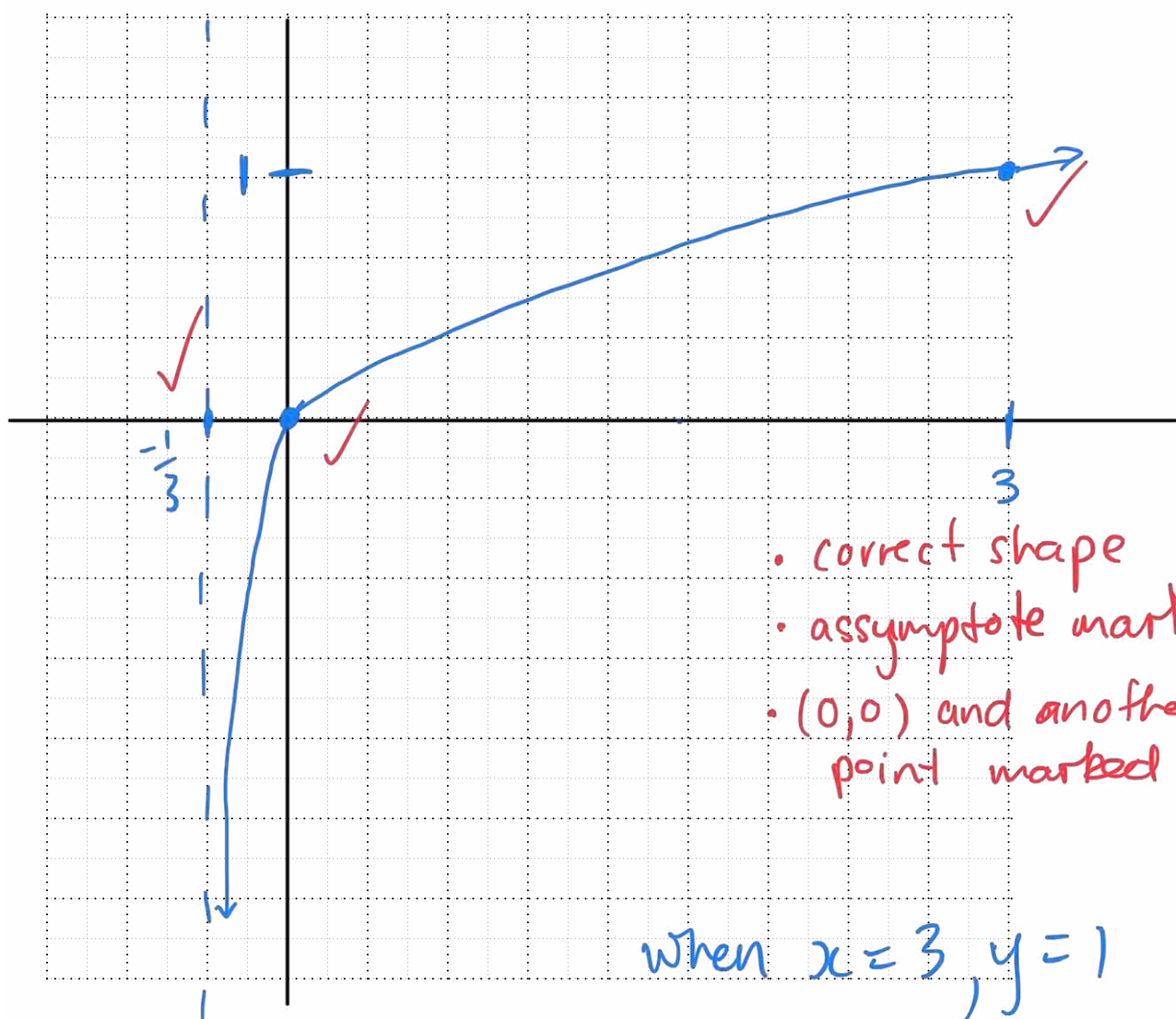
let $x = 0$: $y = \log_{10}(1) = 0$ ✓

2

so $f(x)$ intersects the axes at $(0,0)$ ✓

- (c) Sketch the graph of $f(x)$, showing all important features.

3

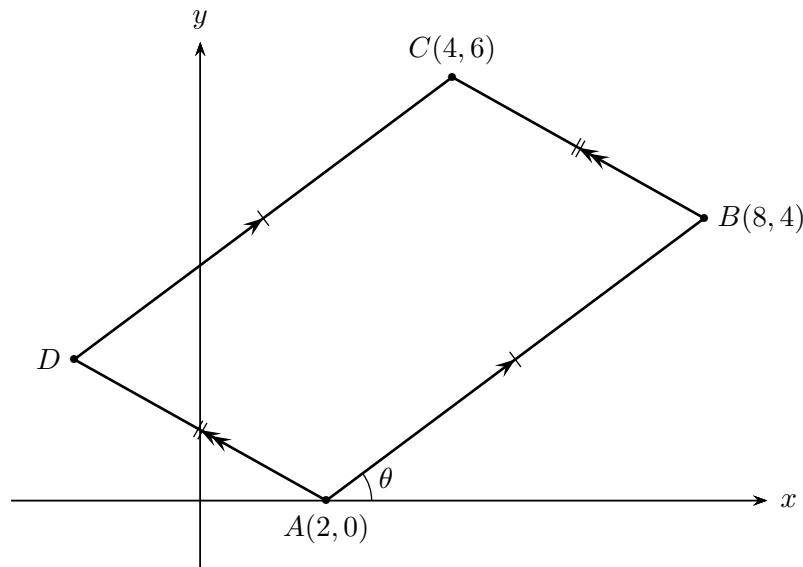


$$3x + 1 = 10$$

$$3x = 9 \quad x = 3$$

Question 7 (6 marks)

The points $A(2, 0)$, $B(8, 4)$, $C(4, 6)$ and D form the four vertices of a parallelogram, as shown in the diagram.



- (a) Find the gradient of the line AB and hence find the angle θ correct to the nearest minute. 2

$$m = \frac{4-0}{8-2} = \frac{4}{6} = \frac{2}{3} \quad \tan \theta = \frac{2}{3} \quad \theta = 33^\circ 41'$$

- (b) Show that the equation of line CD , parallel to AB and passing through C is given by $2x - 3y + 10 = 0$. 2

$$\begin{aligned} y - 6 &= \frac{2}{3}(x - 4) \\ 3y - 18 &= 2x - 8 \\ 0 &= 2x + 10 - 3y \end{aligned} \quad \rightarrow \quad 2x - 3y + 10 = 0$$

- (c) State the coordinates of point D . 2

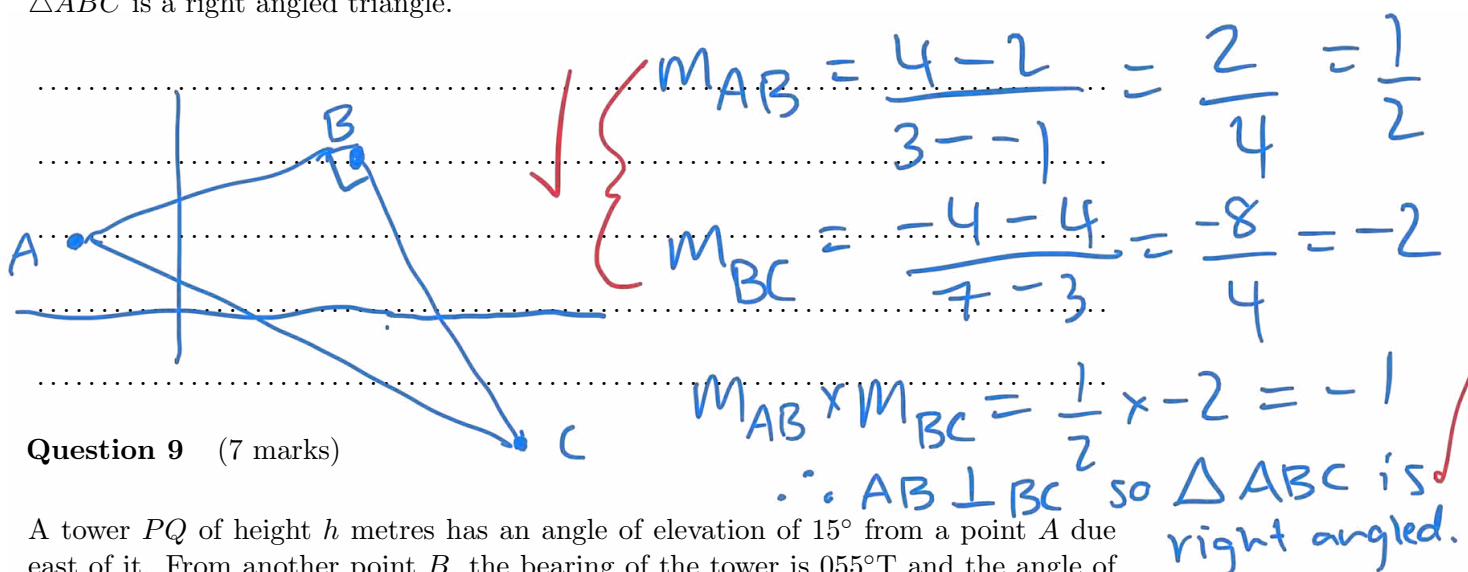
$$\begin{aligned} &\text{across 4, up 2} \\ &\Rightarrow (-2, 2) \end{aligned}$$

Examination continues overleaf...

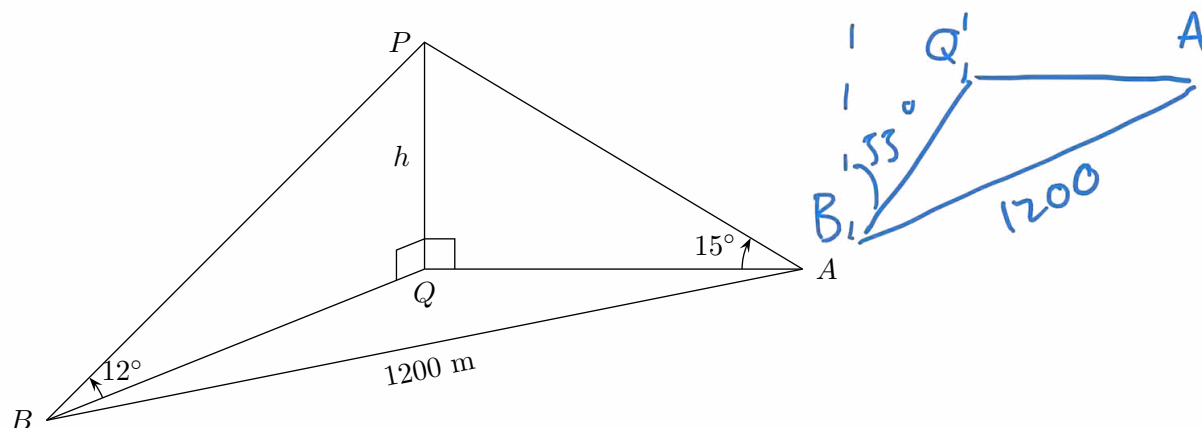
Question 8 (2 marks)

Given the points $A(-1, 2)$, $B(3, 4)$, and $C(7, -4)$ in the Cartesian plane, show that $\triangle ABC$ is a right angled triangle.

2

**Question 9** (7 marks)

A tower PQ of height h metres has an angle of elevation of 15° from a point A due east of it. From another point B , the bearing of the tower is $055^\circ T$ and the angle of elevation is 12° . The points A and B are 1200 metres apart on the same level as the tower's base Q .



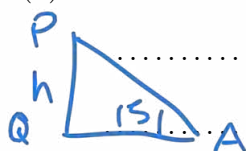
- (a) Show that $\angle AQB = 145^\circ$

$\text{reflex } \angle AQB = 90 + (180 - 55) = 215^\circ$
 $\therefore \angle AQB = 360 - 215 = 145^\circ$

1

- (b) Consider the triangle APQ and show that $AQ = h \cot 15^\circ$.

1



$\cot(15) = \frac{AQ}{h} \Rightarrow AQ = h \cot(15)$

- (c) Find a similar expression for BQ .

1

$BQ = h \cot(12)$

- (d) Show that $h = \frac{1200^2}{\cot 15^\circ \cot 12^\circ - 2 \cot 15^\circ \cot 12^\circ \cos 145^\circ}$

3

cosine rule : $1200^2 = AQ^2 + BQ^2 - 2AQBQ \cos AQB$
 $\triangle ABQ$
 $1200^2 = h^2 \cot^2 15 + h^2 \cot^2 12 - 2h^2 \cot 15 \cot 12$
 $1200^2 = h^2 (\cot^2 15 + \cot^2 12 - 2 \cot 15 \cot 12)$ ✓

$$h^2 = \frac{1200^2}{\cot^2 15 + \cot^2 12 - 2 \cot 15 \cot 12}$$
 ✓

- (e) Hence, calculate the value of h , correct to the nearest metre.

1

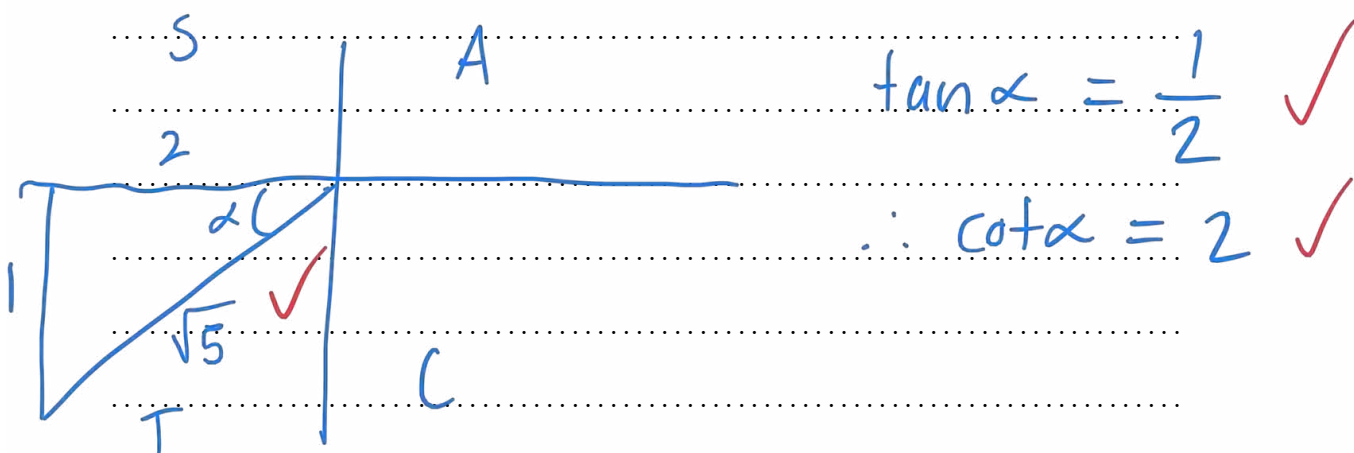
put in calculator : $h = 149.0402 \dots$
 $\approx 149 \text{ m}$ ✓

Question 10 (3 marks)

3

Given that $\cos \alpha = -\frac{2}{\sqrt{5}}$ and $180^\circ \leq \alpha \leq 360^\circ$, find the exact value of $\cot \alpha$.

Show all relevant working.



Examination continues overleaf...

Question 11 (4 marks)

A newly bought car is worth \$10 000. After t years, the value of the car (in dollars) is given by the formula:

$$V = Ae^{-0.2t}$$

- (a) Find the value of A .

1

when $t = 0$, $V = 10\ 000$
 $10\ 000 = Ae^0 \Rightarrow A = 10\ 000$ ✓

- (b) Find the value of the car after 6 years, correct to the nearest cent.

1

$V = 10\ 000 e^{-0.2 \times 6} = \$3\ 011.94$ ✓

- (c) How many years (correct to 2 decimal places) will it take for the car's value to reach \$1 000?

2

$1000 = 10\ 000 e^{-0.2t}$ ✓
 $e^{-0.2t} = 0.1$
 $-0.2t = \ln(0.1)$
 $t = \frac{\ln(0.1)}{-0.2} = 11.51292 \dots$
 $= 11.51 \text{ years}$ ✓

Question 12 (3 marks)

Express:

$$\log(x^2 \sqrt{y^3 z})$$

in the form of $a \log x + b \log y + c \log z$.

3

$\log(x^2 y^{\frac{3}{2}} z^{\frac{1}{2}})$ ✓
 $= \log x^2 + \log y^{\frac{3}{2}} + \log z^{\frac{1}{2}}$ ✓
 $= 2 \log x + \frac{3}{2} \log y + \frac{1}{2} \log z$ ✓

End of paper.